

Wireless components. Most toys these days include wireless transceivers. 2.4 GHz wireless capability is not uncommon.

Actuators. Motors are the most common output of any toy. These may differ from basic direct-current (DC) motors to servo motors, depending on their application. "Some toys have IC engines on them, whereas for game-stations, the output is graphics on a television or computer screen," explains Azad.

Selection of components

Most of the components used in today's toys seem like those used in robots and smartphones, yet one cannot afford to use the very same high-end equipment in toys as in robots all the time. "In the selection of components, cost is one of the deciding factors. With similar set of features and functions, a toy cannot be priced as high as a robot. Since the sales volume is higher for



A programmable, radio-controlled model aircraft

toys, this issue can be handled to some extent," says Jayakrishnan.

Some of the components are custom-made by or for the toymakers, while at other times the toymakers simply use off-the-shelf components. "The Lego Mindstorm processor, for example, is custom-made as the pro-

cessor didn't exist. Besides this, the Xbox 360 and Konnect sensors are also completely custom-made using existing technology. For some other simple autonomous applications, ready-made sensors and processes are also used," says Azad.

Off-the-shelf components reduce the time and cost of development and are the first choice of any company. The cost of development of custom-made devices is quite high. A company would go in for it only when they are sure that it is going to be a disruptive technology, which will ensure them huge profits. While this difference is less critical for external and mechanical components, the cost difference can be extremely high for electronic components.

"For embedded platforms and sensors, toys and actual robots share a lot of common components. To cut down the cost companies often go for a different sort of package. Some chips can be bought as dies and then can be directly wire-bound to a PCB with an epoxy covering. In any case, the chip package can contribute up to near 50 per cent of the cost," opines Jayakrishnan.

Toy-making is not child's play

Forget being child's play, it is not even just an art. Toy-making is a complete science. "It might comprise embedded

A detailed example—the Genibo-QD Robot Dog

Genibo is an articulated dog-like four-legged robot that can do all kinds of motions including scratching the ground, standing on front paws, dancing, taekwondo, etc. Genibo expresses its feelings through approximately 700 emotional actions, the emotion LEDs in its eyes, and puppy sounds. It can recognise human faces, such as the owner's face through the camera on its nose. As it gets closer to its owner, the Genibo becomes happier and better at finding and following its owner.

It also has eleven touch sensors, a speaker, voice and visual recognition. The gyroscope, IR interface and remote controller can be used for a more sophisticated control of actions, sounds and emoticons that assist in communication.

Some specifications of the Robot Dog are as follows:

Model number	Genibo-QD
Number of Legs	4
Sensors	➤ Touch sensors: human touch sensors ➤ Inclinator sensor: two-axis sensor ➤ Floor detection sensor: push-button (paws) ➤ Distance measurement sensor: PSD ➤ Recognition distance (camera): 10~80cm ➤ Microphone: voice recognition, 100 words ➤ Camera: 1.3 MP, 0.64cm (0.25-inch) CMOS ➤ Display: back-touch LED, three-coloured eye LED
Servos	DC servos (15 joints)
Power	DC 9V, 2.5A
Processor	32-bit, 350MHz RISC processor
Operating system	Linux (embedded), Windows
Wireless	Infrared controller and 802.11b/g

—N.H. Shekar